CHAPTER 11

The Muscular System

Gross anatomy of skeletal muscle — interactions, naming, and the axial and appendicular muscles of the body.

Anatomy & Physiology · Broward-Miami Health Institute

FACULTY EDITION

Learning Objectives

After studying this chapter, students will be able to:

1. Describe the actions and roles of agonists and antagonists
2. Explain the structure of muscle fascicles and their role in generating force
3. Explain the criteria used to name skeletal muscles
4. Identify the major skeletal muscles and their actions on the body
5. Identify the origins and insertions of muscles and the prime movers

Movement is the product of muscle, bone, and nervous control working together.

11.1

SECTION

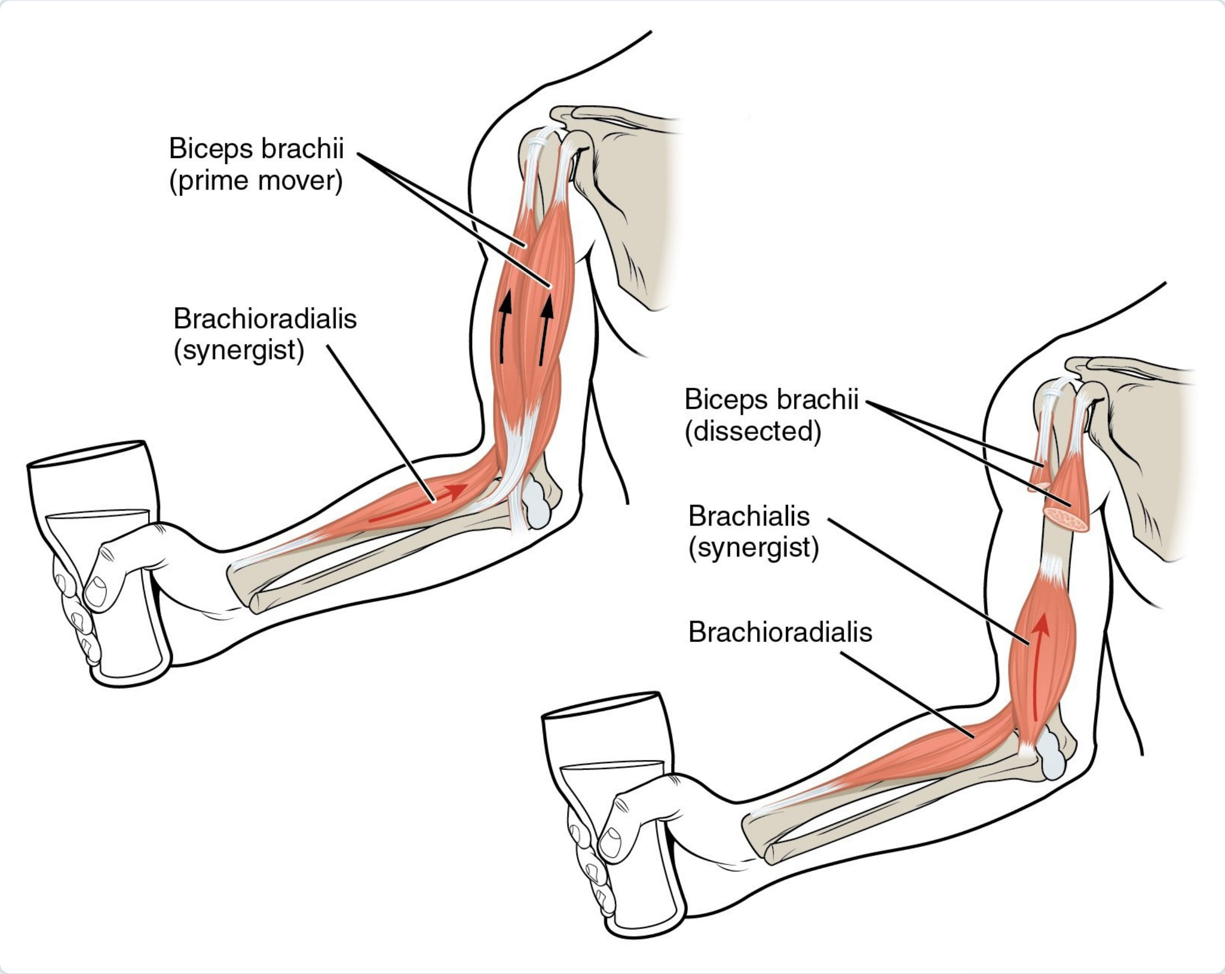
Interactions of Skeletal Muscles, Fascicle Arrangement & Lever Systems

How muscles work in groups, how fascicle geometry shapes force, and how bones act as levers.

SECTION 11.1

Agonists, Antagonists & Synergists

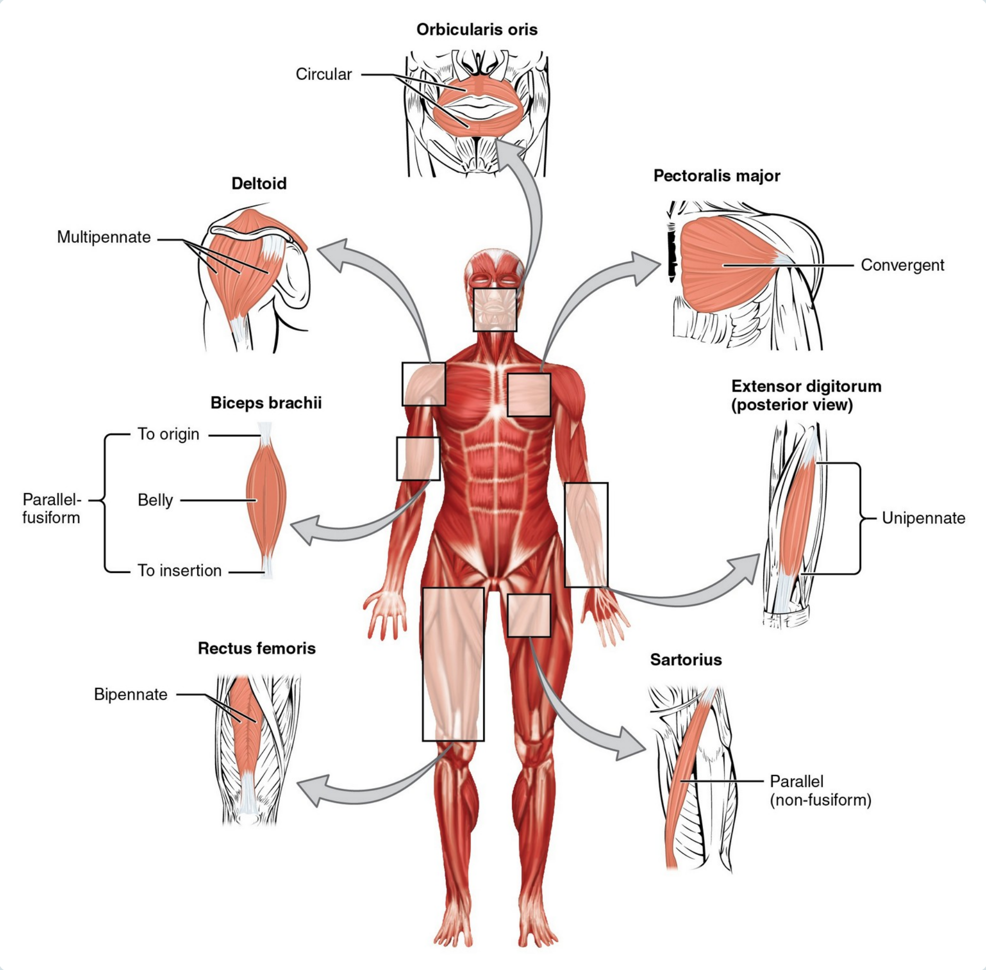
- Prime mover (agonist) produces the main movement
- Antagonist opposes and controls the motion
- Synergists assist; fixators stabilize the origin
- Example: biceps flexes, triceps extends the elbow



Biceps brachii acting as a prime mover with synergists

Fascicle Arrangement & Muscle Shapes

- Fascicle pattern sets the force–range trade-off
- Parallel, fusiform, convergent, circular, pennate
- Pennate muscles pack more fibers → greater force
- Circular muscles (sphincters) close body openings



Patterns of fascicle organization seen throughout the body

SECTION 11.1

Lever Systems & Muscle Action

- Bones are levers; joints are fulcrums; muscle is effort
- First-, second-, and third-class levers occur in the body
- Most body levers are third-class (favor speed/range)
- Leverage lets a small effort move a larger load

Muscle generates movement by pulling on bony levers

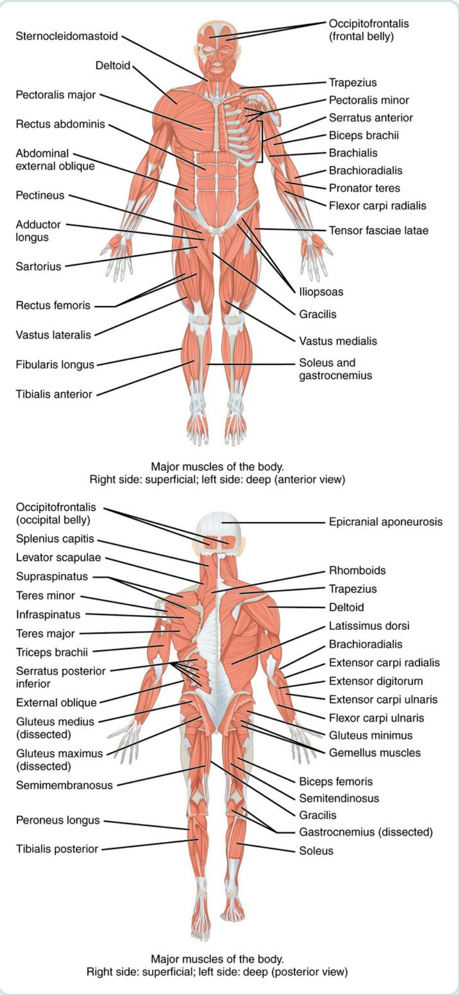
11.2

SECTION

Naming Skeletal Muscles

Muscle names are descriptive codes — decode them to predict location and action.

How Skeletal Muscles Are Named



Major muscles of the body.
Right side: superficial; left side: deep (anterior view)

Major muscles of the body.
Right side: superficial; left side: deep (posterior view)

Major skeletal muscles, anterior and posterior views

- Named by location, shape, size, and fiber direction
- Also by number of origins and by origin/insertion
- Named by action: flexor, extensor, adductor
- Decoding the name predicts what the muscle does

Naming Criteria — Worked Examples

Example	Word	Latin Root 1	Latin Root 2	Meaning	Translation
abductor digiti minimi	abductor	ab = away from	duct = to move	a muscle that moves away from	A muscle that moves the little finger or toe away
	digiti	digitus = digit		refers to a finger or toe	
	minimi	minimus = mini, tiny		little	
adductor digiti minimi	adductor	ad = to, toward	duct = to move	a muscle that moves towards	A muscle that moves the little finger or toe toward
	digiti	digitus = digit		refers to a finger or toe	
	minimi	minimus = mini, tiny		little	

- Break each name into root + meaning
- "Abductor digiti minimi" → moves the little finger
- Build a personal glossary of recurring roots
- Use the name to predict location and action

Reference: word roots used in muscle names

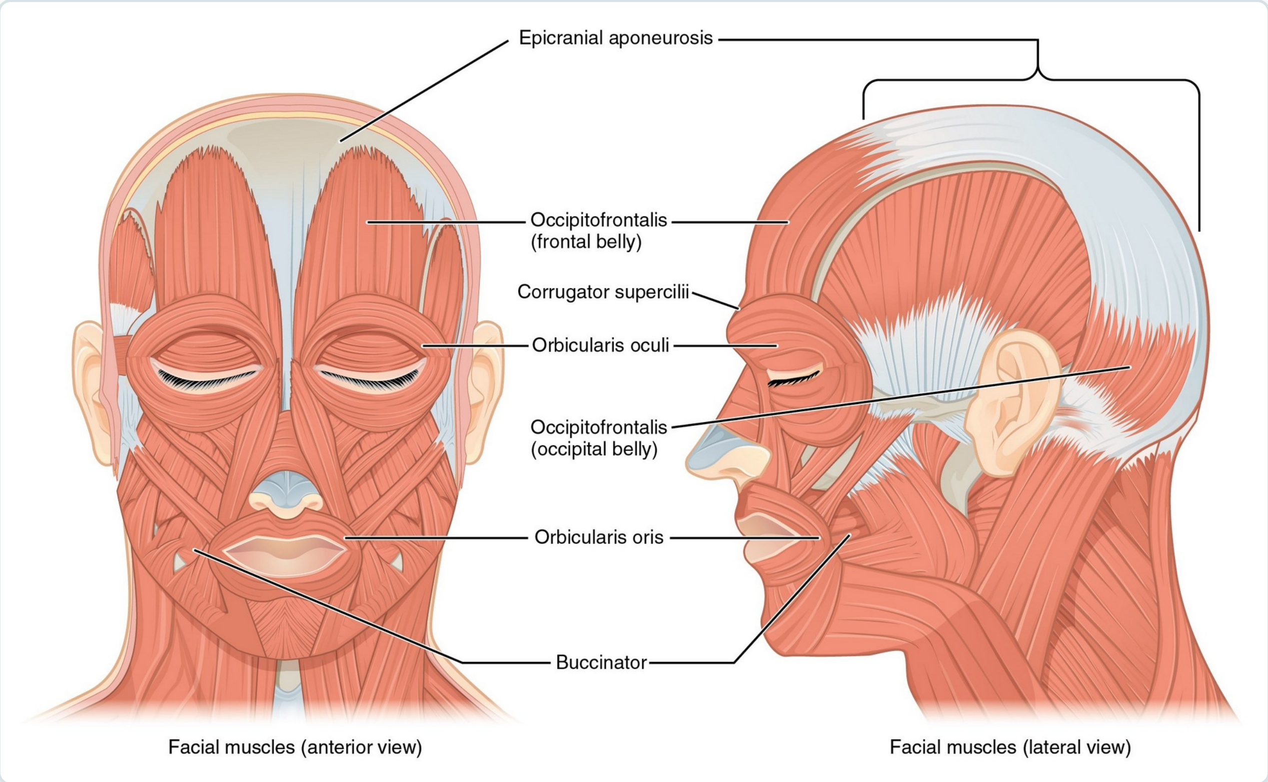
11.3

SECTION

Axial Muscles of the Head, Neck & Back

Facial expression, mastication, eye and tongue movement, neck support, and spinal posture.

Muscles of Facial Expression



Facial muscles, anterior and lateral views

- Insert into skin, not bone → move soft tissue
- All innervated by the facial nerve (CN VII)
- Orbicularis oris/oculi, zygomaticus, buccinator
- Clinical: CN VII palsy causes facial drooping

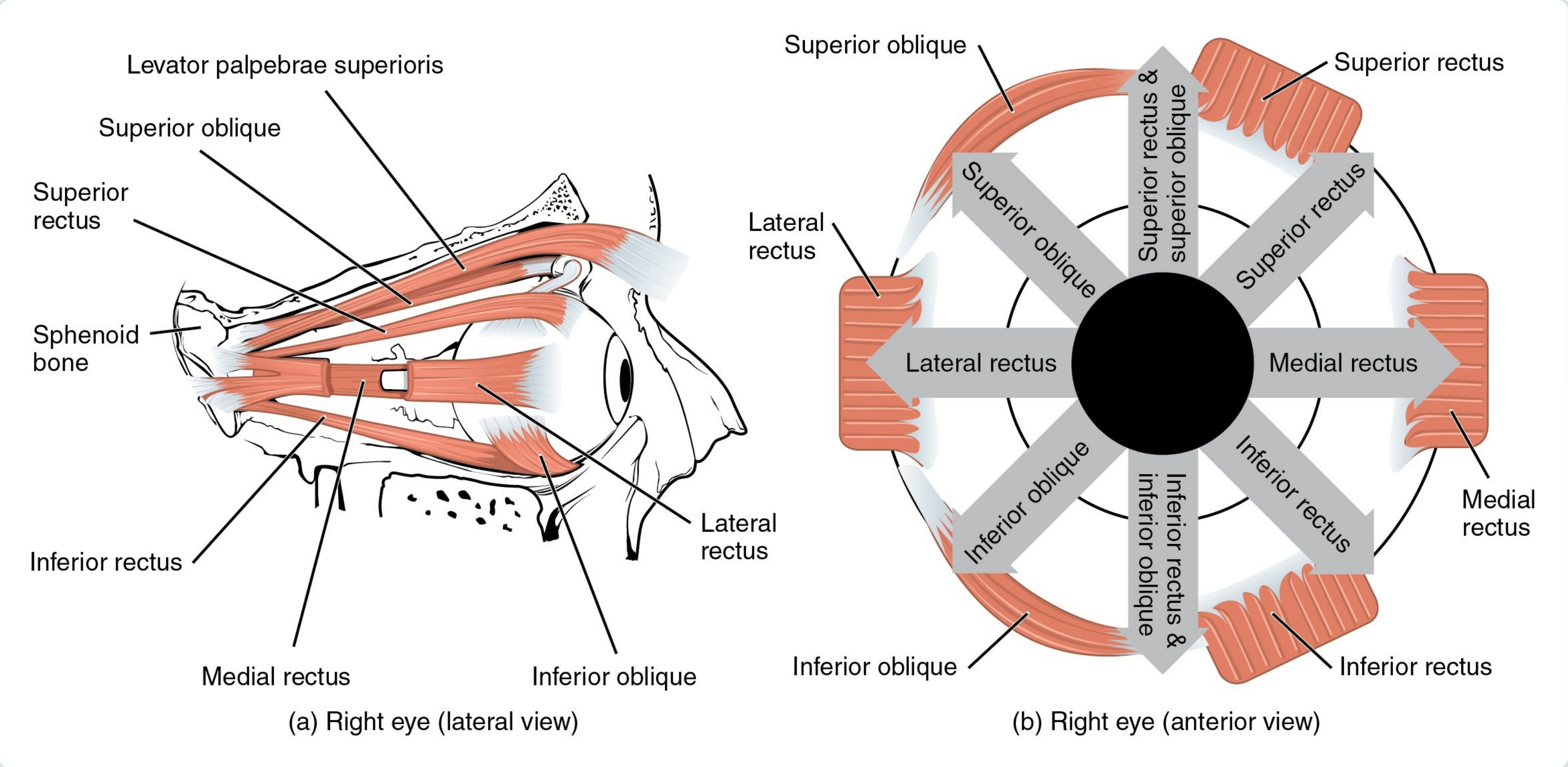
Facial Expression — Reference Table

Movement	Target	Target motion direction	Prime mover	Origin	Insertion
Brow					
Raising eyebrows (e.g., showing surprise)	Skin of scalp	Anterior	Occipito-frontalis, frontal belly	Epicraneal aponeurosis	Underneath skin of forehead
Tensing and retracting scalp	Skin of scalp	Posterior	Occipito-frontalis, occipital belly	Occipital bone; mastoid process (temporal bone)	Epicraneal aponeurosis
Lowering eyebrows (e.g., scowling, frowning)	Skin underneath eyebrows	Inferior	Corrugator supercilii	Frontal bone	Skin underneath eyebrow
Nose					
Flaring nostrils	Nasal cartilage (pushes nostrils open when cartilage is compressed)	Inferior compression; posterior compression	Nasalis	Maxilla	Nasal bone
Mouth					
Raising upper lip	Upper lip	Elevation	Levator labii superioris	Maxilla	Underneath skin at corners of the mouth; orbicularis oris
Lowering lower lip	Lower lip	Depression	Depressor labii inferioris	Mandible	Underneath skin of lower lip
Opening mouth and sliding lower jaw left and right	Lower jaw	Depression, lateral	Depressor angulus oris	Mandible	Underneath skin at corners of mouth
Smiling	Corners of mouth	Lateral elevation	Zygomaticus major	Zygomatic bone	Underneath skin at corners of mouth (dimple area); orbicularis oris
Shaping of lips (as during speech)	Lips	Multiple	Orbicularis oris	Tissue surrounding lips	Underneath skin at corners of the mouth
Lateral movement of cheeks (e.g., sucking on a straw; also used to compress air in mouth while blowing)	Cheeks	Lateral	Buccinator	Maxilla, mandible; sphenoid bone (via pterygomandibular raphae)	Orbicularis oris
Pursing of lips by straightening them laterally	Corners of mouth	Lateral	Risorius	Fascia of parotid salivary gland	Underneath skin at corners of the mouth
Protrusion of lower lip (e.g., pouting expression)	Lower lip and skin of chin	Protraction	Mentalis	Mandible	Underneath skin of chin

Muscles of facial expression: action, origin, insertion

- Group muscles by region: eye, mouth, nose, scalp
- Note the shared CN VII innervation
- Focus on action verbs: elevate, compress, purse
- Pair with the diagram on the previous slide

Muscles of the Eye & Mastication

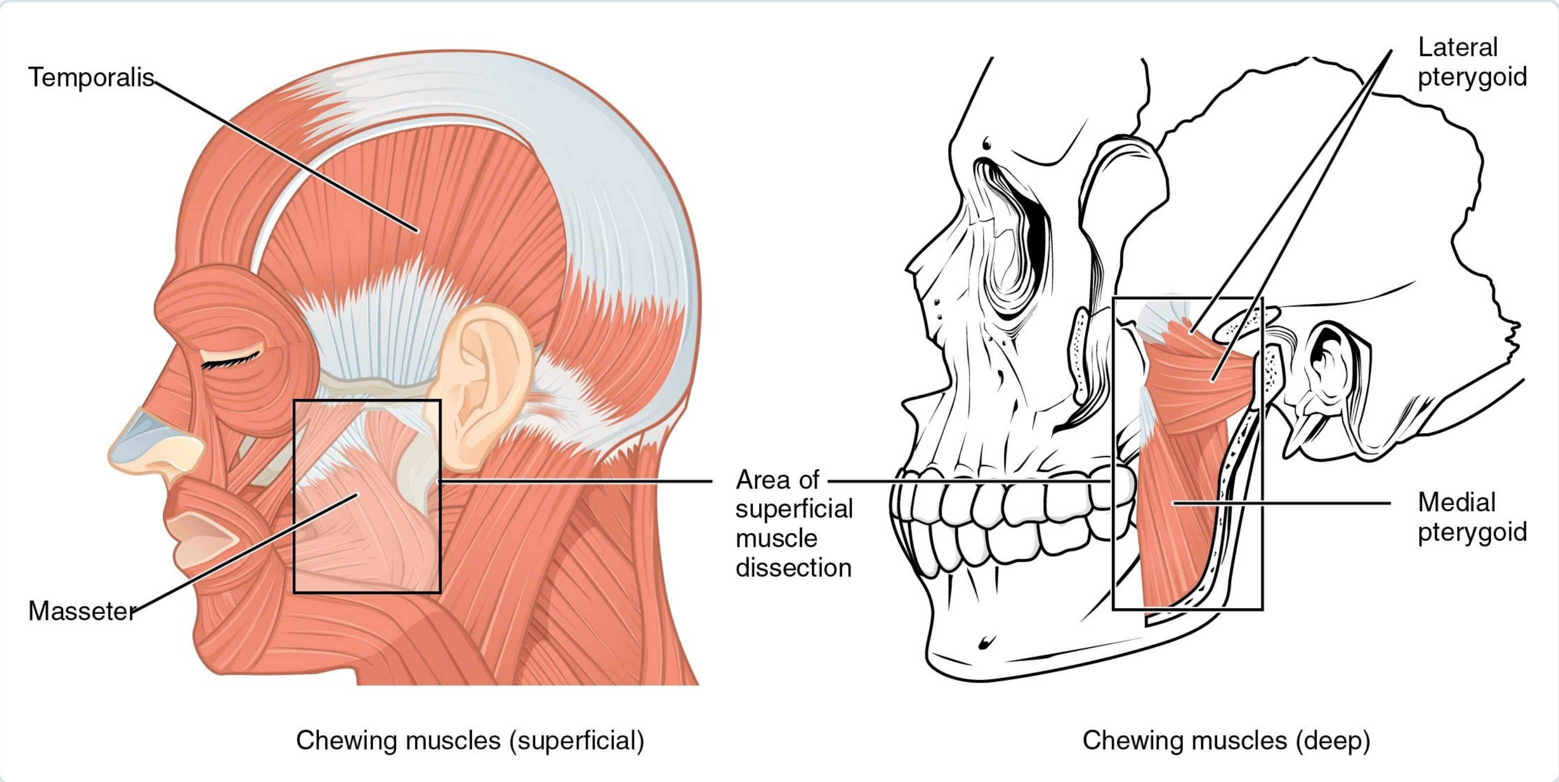


Extraocular muscles and the muscles of mastication

- Six extraocular muscles move the eyeball
- Mastication: masseter, temporalis, pterygoids
- Chewing muscles served by trigeminal nerve (CN V)
- Clinical: check jaw strength and symmetry

SECTION 11.3

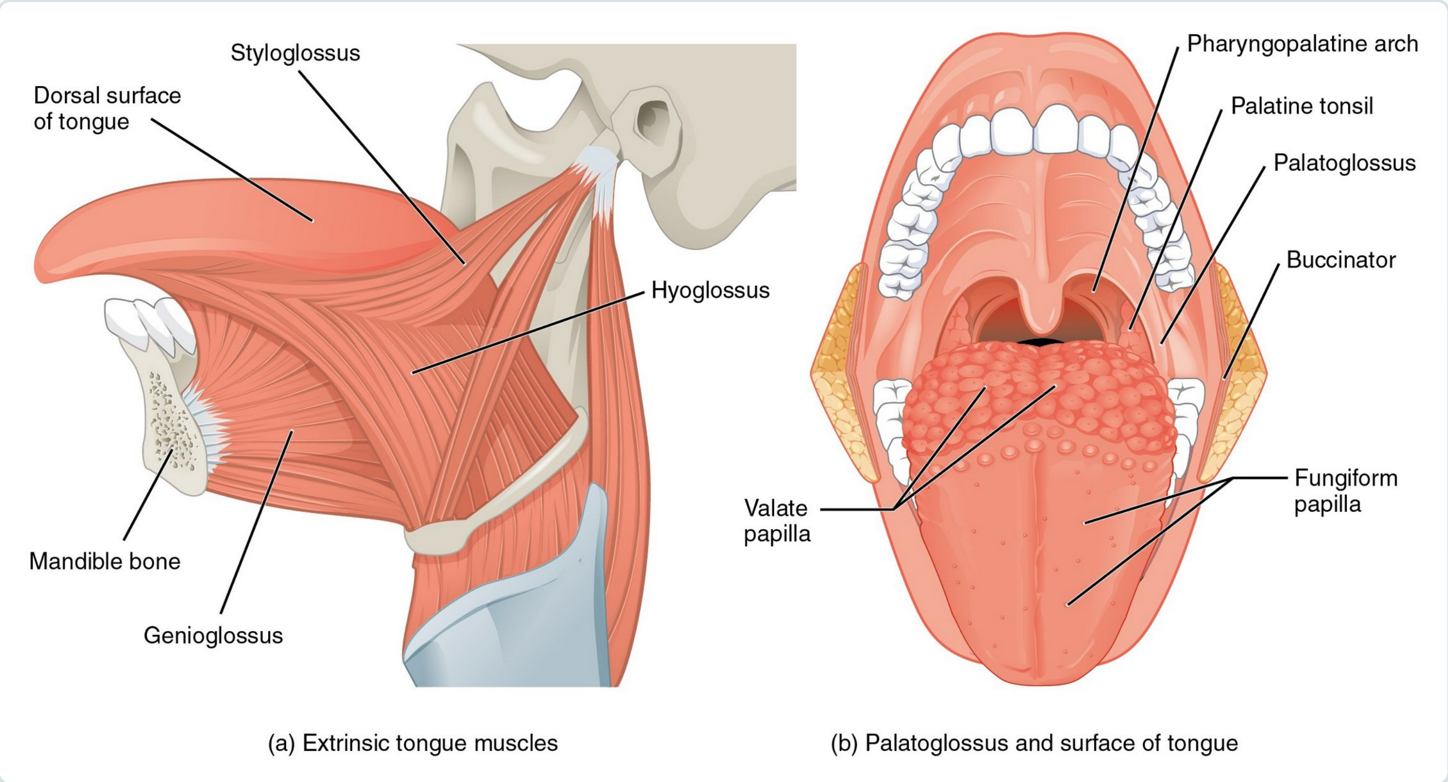
Muscles That Move the Tongue



Muscles of mastication and the tongue

- Intrinsic muscles change the tongue's shape
- Extrinsic muscles (-glossus) move its position
- Innervated by the hypoglossal nerve (CN XII)
- Essential for speech, chewing, and swallowing

Tongue & Pharyngeal Muscles



Tongue muscles and associated pharyngeal structures

- Coordinate the oral phase of swallowing
- Move the food bolus toward the pharynx
- Work with suprahyoid muscles to elevate the larynx
- Dysfunction → dysphagia and aspiration risk

SECTION 11.3

Head & Neck — Reference Table

Movement	Target	Target motion direction	Prime mover	Origin	Insertion
Tongue					
Moves tongue down; sticks tongue out of mouth	Tongue	Inferior (depresses); anterior (protracts)	Genioglossus	Mandible	Tongue undersurface; hyoid bone
Moves tongue up; retracts tongue back into mouth	Tongue	Superior (elevates); posterior (retracts)	Styloglossus	Temporal bone (styloid process)	Tongue undersurface and sides
Flattens tongue	Tongue	Inferior (depresses)	Hyoglossus	Hyoid bone	Sides of tongue
Bulges tongue	Tongue	Superior (elevation)	Palatoglossus	Soft palate	Side of tongue
Swallowing and speaking					
Raises the hyoid bone in a way that also raises the larynx, allowing the epiglottis to cover the glottis during deglutition; also assists in opening the mouth by depressing the mandible	Hyoid bone; larynx	Superior (elevates)	Digastric	Mandible; temporal bone	Hyoid bone
Raises and retracts the hyoid bone in a way that elongates the oral cavity during deglutition	Hyoid bone	Superior (elevates); posterior (retracts)	Stylohyoid	Temporal bone (styloid process)	Hyoid bone
Raises hyoid bone in a way that presses tongue against the roof of the mouth, pushing food back into the pharynx during deglutition	Hyoid bone	Superior (elevates)	Mylohyoid	Mandible	Hyoid bone; median raphe
Raises and moves hyoid bone forward, widening pharynx during deglutition	Hyoid bone	Superior (elevates); anterior (protracts)	Geniohyoid	Mandible	Hyoid bone
Retracts hyoid bone and moves it down during later phases of deglutition	Hyoid bone	Inferior (depresses); posterior (retracts)	Omohyoid	Scapula	Hyoid bone
Depresses the hyoid bone during swallowing and speaking	Hyoid bone	Inferior (depresses)	Sternohyoid	Clavicle	Hyoid bone
Shrinks distance between thyroid cartilage and hyoid bone, allowing production of high-pitch vocalizations	Hyoid bone; thyroid cartilage	Hyoid bone: inferior (depresses); thyroid cartilage: superior (elevates)	Thyrohyoid	Thyroid cartilage	Hyoid bone
Depresses larynx, thyroid cartilage, and hyoid bone to create different vocal tones	Larynx; thyroid cartilage; hyoid bone	Inferior (depresses)	Sternothyroid	Sternum	Thyroid cartilage
Rotates and tilts head to he side; tilts head forward	Skull; cervical vertebrae	Individually: medial rotation; lateral flexion; bilaterally: anterior (flexes)	Sternocleid-omastoid; semispinalis capitis	Sternum; clavicle	Temporal bone (mastoid process); occipital bone
Rotates and tilts head to the side; tilts head backwards	Skull; cervical vertebrae	Individually: lateral rotation; lateral flexion; bilaterally: anterior (flexes)	Splenius capitis; longissimus capitis		

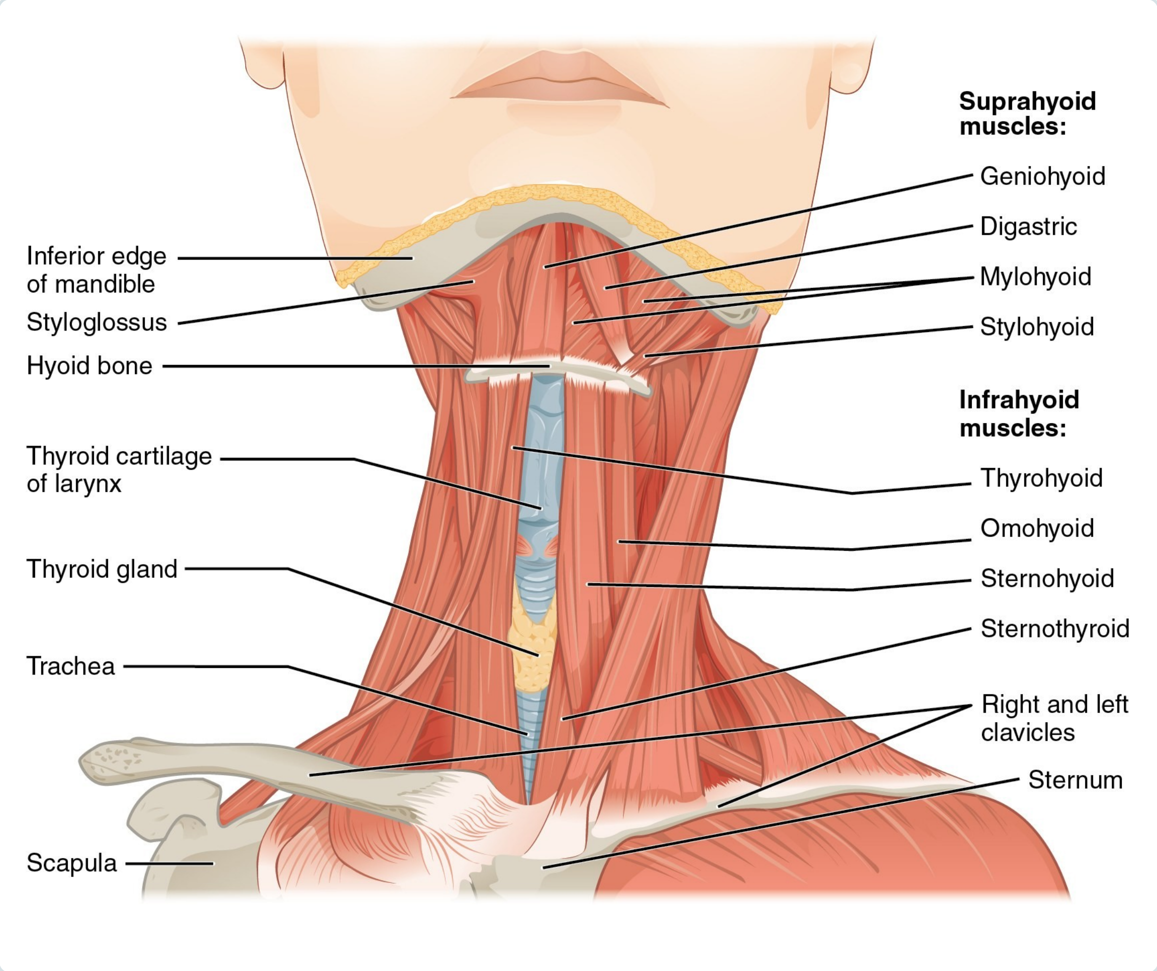
Muscles of the head and neck: summary table

- Consolidates head and neck muscle actions
- Cross-reference each muscle's cranial-nerve supply
- Identify the prime mover for each motion
- Use for targeted review before exams

SECTION 11.3

Anterior Neck — Hyoid Muscles

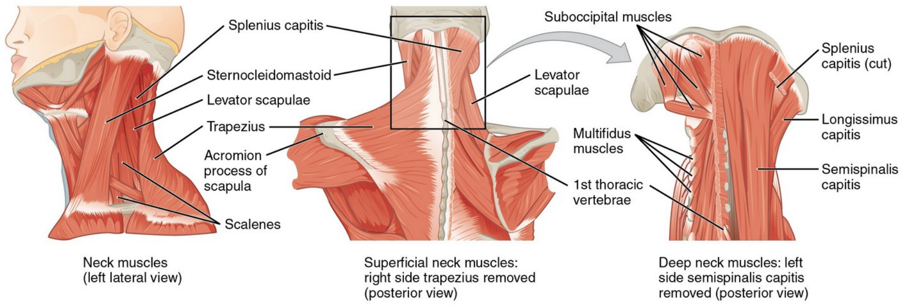
- Suprahyoid muscles elevate the hyoid and larynx
- Infrahyoid (strap) muscles depress them
- Stabilize the hyoid for speech and swallowing
- Landmarks for airway and thyroid assessment



Suprahyoid and infrahyoid muscles of the neck

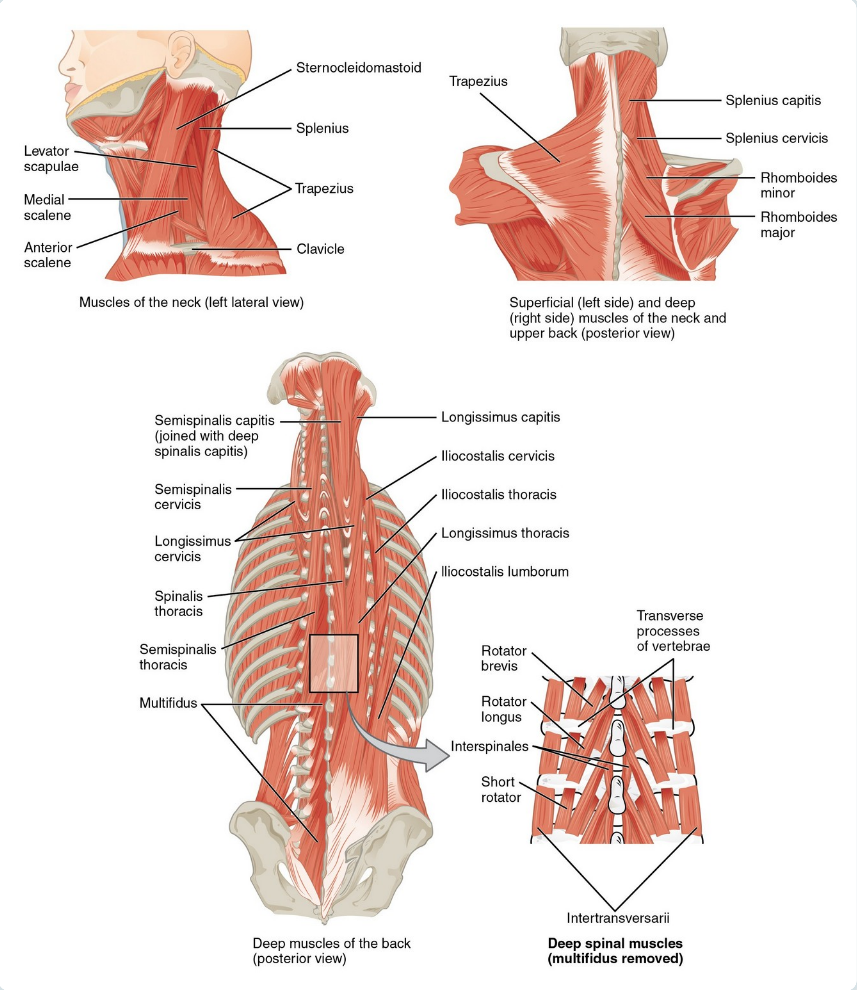
Muscles of the Neck & Back

- Sternocleidomastoid flexes and rotates the head
- Trapezius and scalenes support posture
- Deep muscles stabilize the cervical spine
- Scalenes act as accessory respiratory muscles



Posterior and lateral muscles of the neck and back

Deep Back — Erector Spinae



Erector spinae group and the deep back muscles

- Erector spinae = iliocostalis, longissimus, spinalis
- Extend and stabilize the vertebral column
- Maintain upright posture against gravity
- Common source of mechanical low-back strain

11.4

SECTION

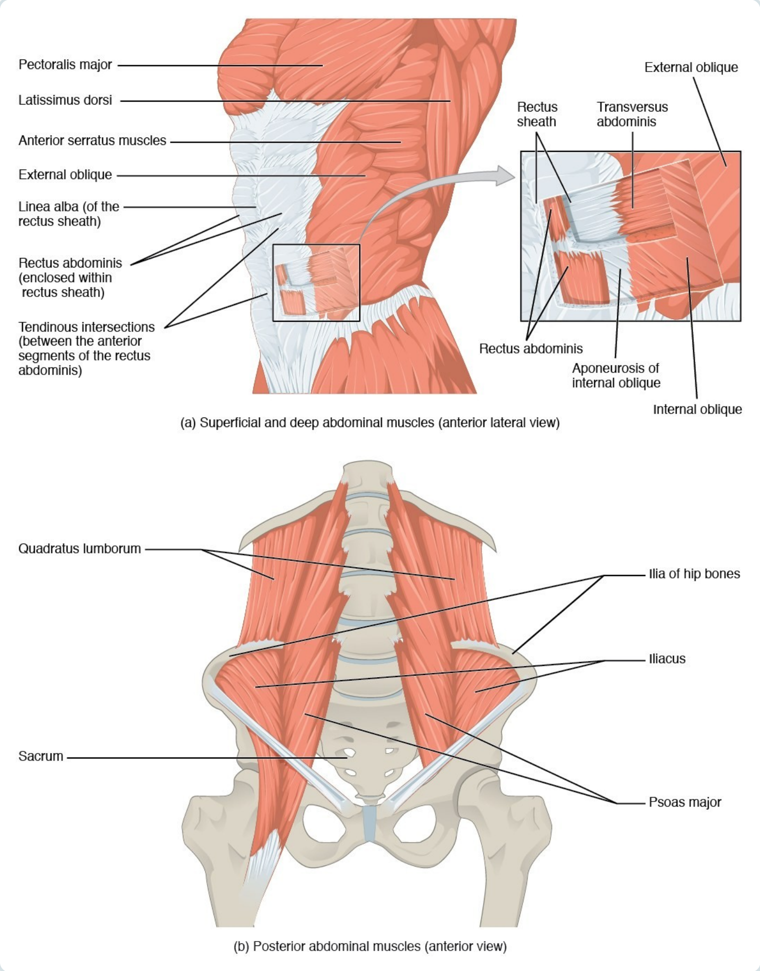
Axial Muscles of the Abdominal Wall & Thorax

The core, the diaphragm and intercostals of breathing, and the pelvic floor.

SECTION 11.4

Muscles of the Abdominal Wall

- Rectus abdominis flexes the trunk
- External/internal obliques bend and rotate
- Transversus abdominis compresses the abdomen
- Support posture, breathing, and core stability

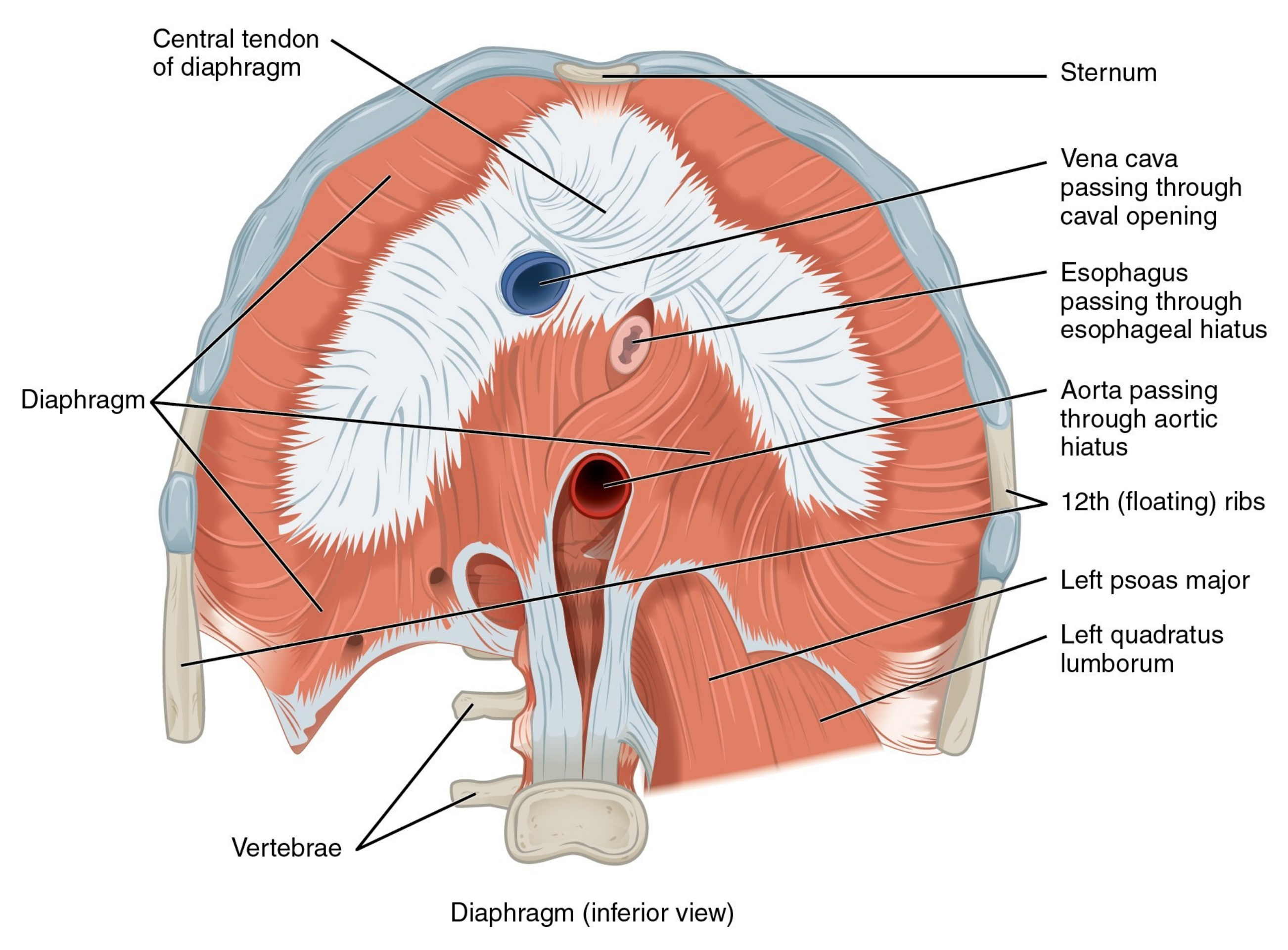


Layered muscles of the anterior abdominal wall

SECTION 11.4

The Diaphragm

- Primary muscle of inspiration
- Contracts and flattens → thoracic volume rises
- Innervated by the phrenic nerve (C3–C5)
- "C3, 4, 5 keep the diaphragm alive"

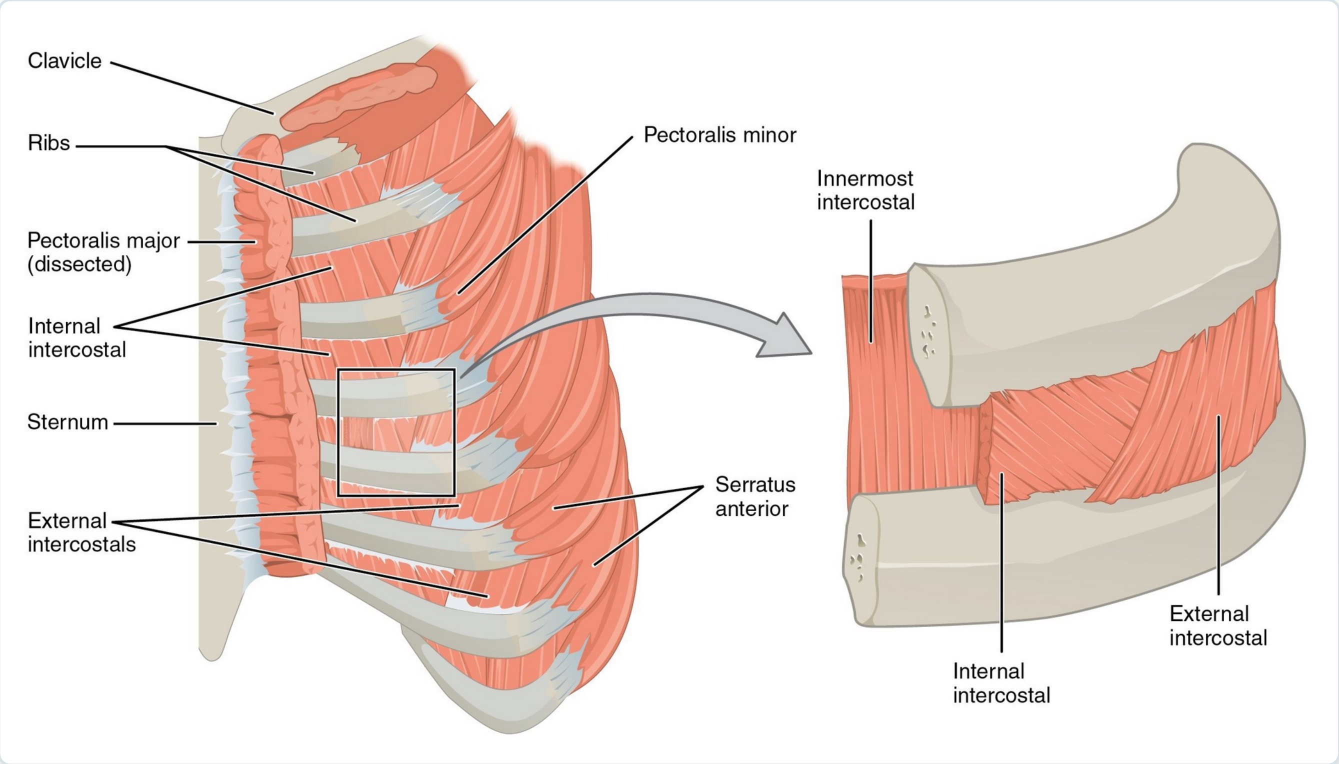


The diaphragm, inferior view

SECTION 11.4

Thoracic Muscles & Intercostals

- External intercostals elevate ribs (inspiration)
- Internal intercostals depress ribs (forced expiration)
- Accessory muscles assist in respiratory distress
- Retractions signal increased work of breathing

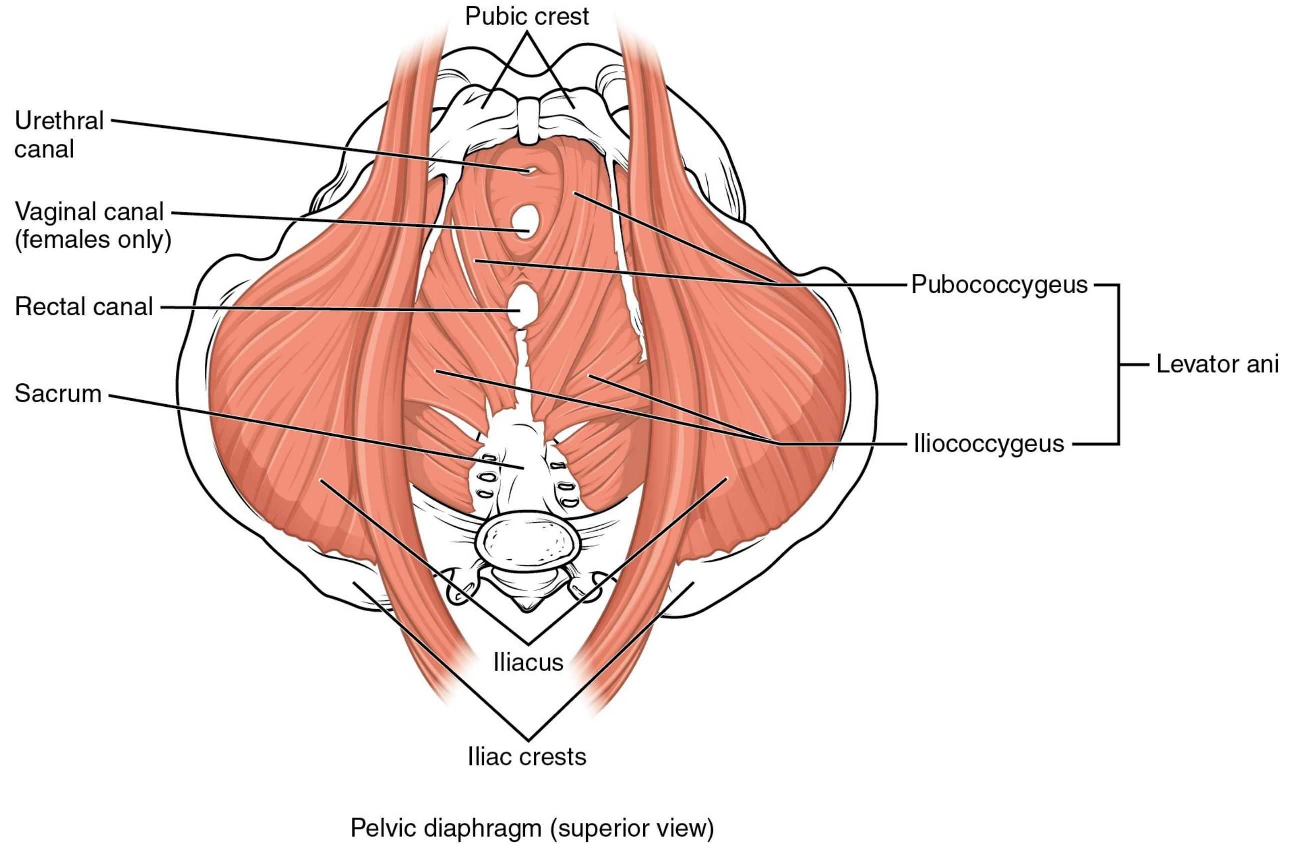


Intercostal muscles of the thoracic wall

SECTION 11.4

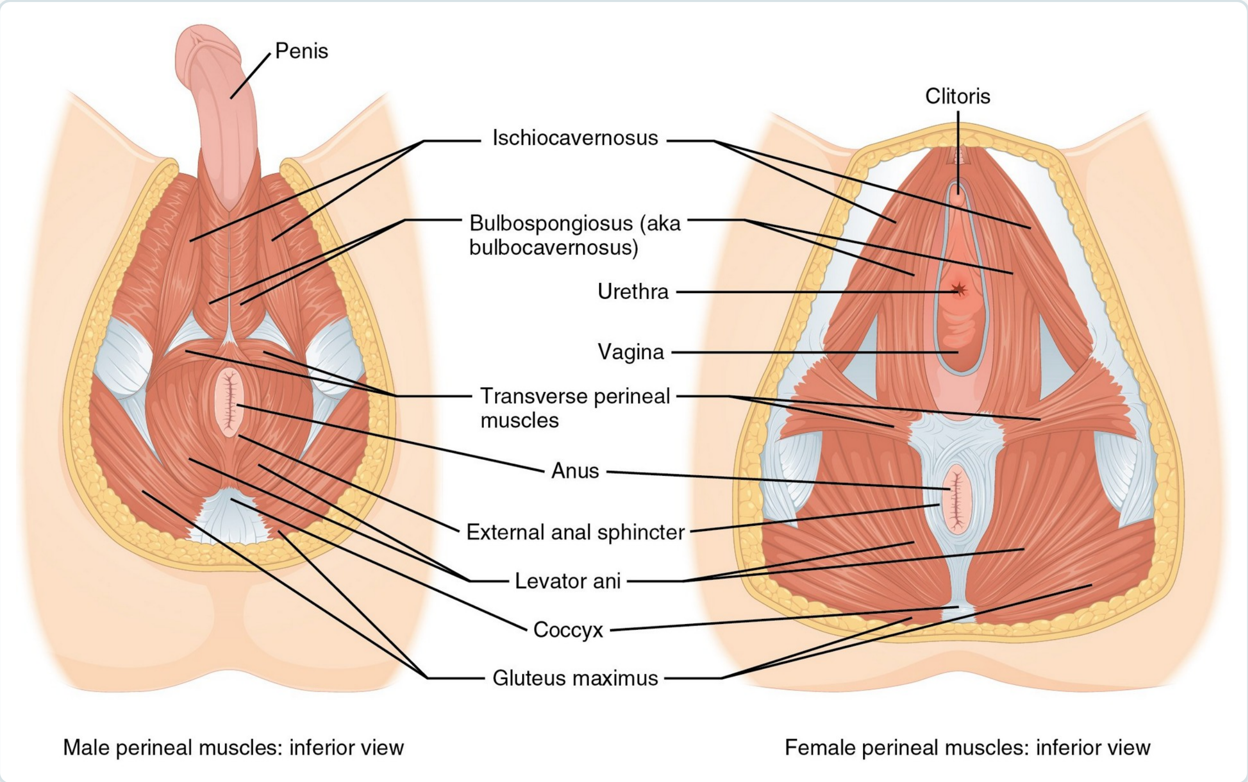
Pelvic Floor & Perineum

- Levator ani forms the pelvic diaphragm
- Supports the bladder, rectum, and pelvic organs
- Maintains continence of urine and stool
- Weakness → prolapse and incontinence



Muscles of the pelvic floor (perineum)

The Pelvic Diaphragm



Pelvic diaphragm, male and female

- Levator ani + coccygeus = pelvic diaphragm
- Sex differences in the openings it transmits
- Resists downward intra-abdominal pressure
- Key to bowel, bladder, and sexual function

11.5

SECTION

Muscles of the Pectoral Girdle & Upper Limbs

Positioning the scapula, moving the humerus and forearm, and the muscles of the hand.

Pectoral Girdle — Reference Table

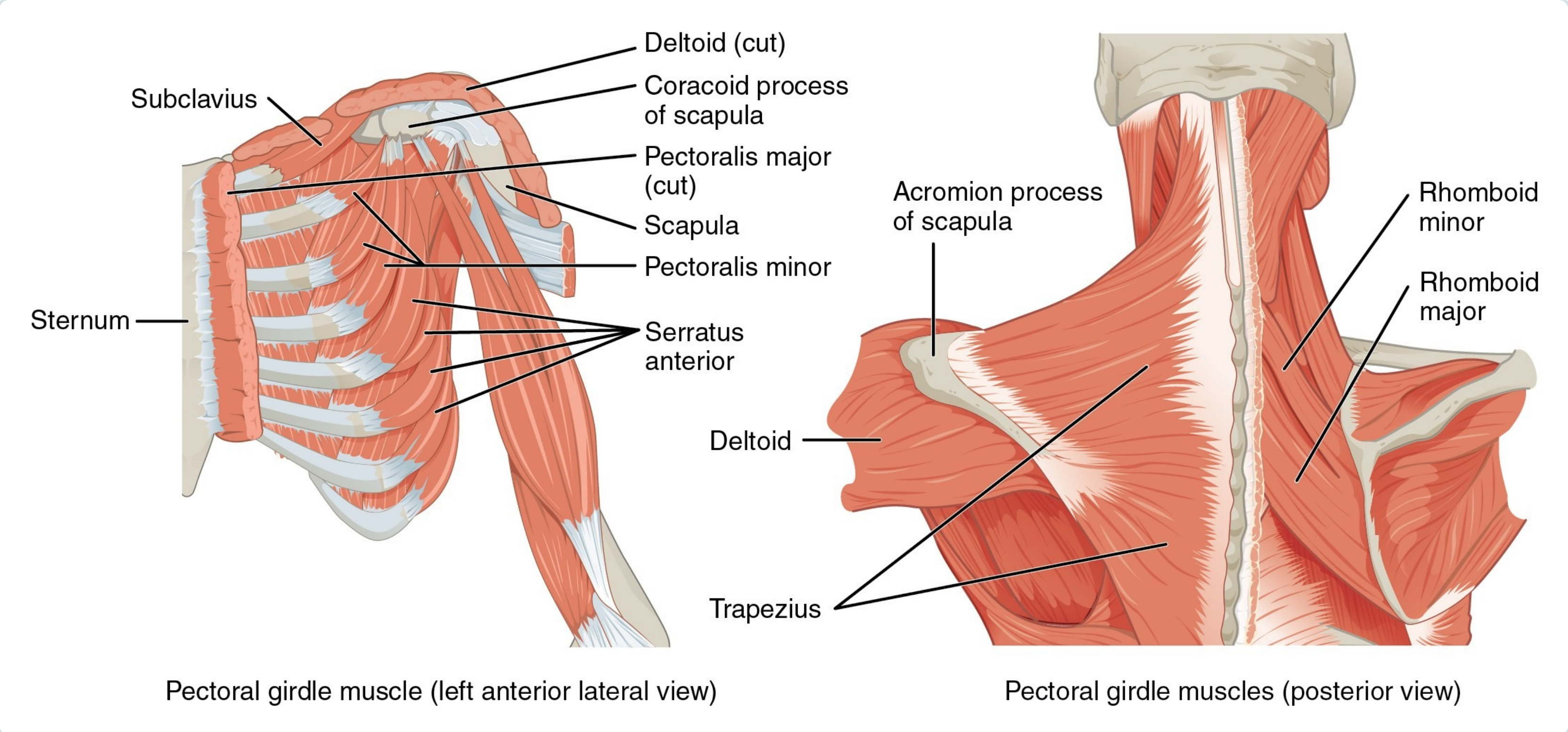
Movement	Target	Target motion direction	Prime mover	Origin	Insertion
Defecation; urination; birth; coughing	Abdominal cavity	Superior (resists pressure during abdominal compression)	Levator ani pubococcygeus; levator ani iliococcygeus	Pubis; ischium	Urethra; anal canal; perineal body; coccyx
Superficial muscles					
None—supports perineal body maintaining anus at center of perineum	Perineal body	None	Superficial transverse perineal	Ischium	Perineal body
Involuntary response that compresses urethra when excreting urine in both sexes or while ejaculating in males; also aids in erection of penis in males	Urethra	Compression	Bulbospongiosus	Perineal body	Perineal membrane; corpus spongiosum of penis; deep fascia of penis; clitoris in female
Compresses veins to maintain erection of penis in males; erection of clitoris in females	Veins of penis and clitoris	Compression	Ischiocavernosus	Ischium; ischial rami; pubic rami	Pubic symphysis; corpus cavernosum of penis in male; clitoris of female
Deep muscles					
Voluntarily compresses urethra during urination	Urethra	Compression	External urethral sphincter	Ischial rami; pubic rami	Male: median raphe; female: vaginal wall
Closes anus	Anus	Sphincter	External anal sphincter	Anococcygeal ligament	Perineal body

Muscles that position the pectoral girdle

- Anchors the upper limb to the axial skeleton
- These muscles position the scapula and clavicle
- A stable scapula enables efficient arm movement
- Sets up the humerus muscles that follow

SECTION 11.5

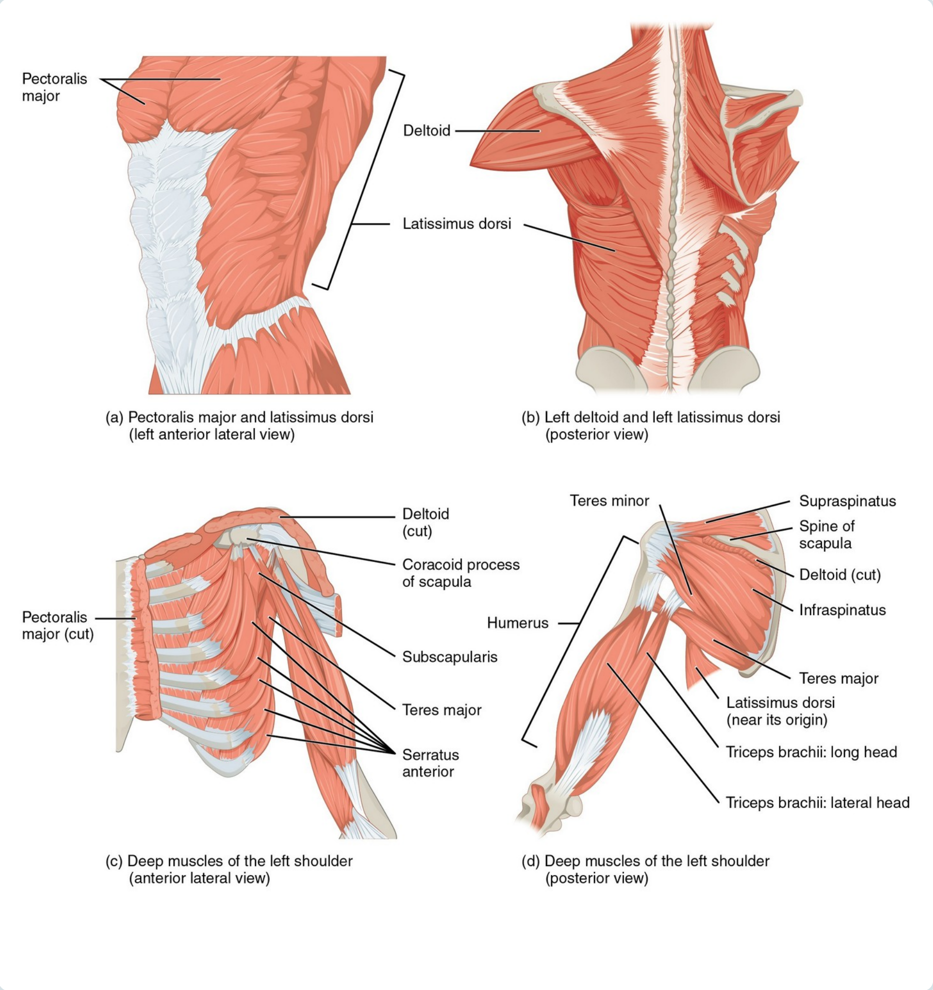
Positioning the Pectoral Girdle



Anterior and posterior muscles of the pectoral girdle

- Trapezius and serratus anterior move the scapula
- Rhomboids retract; pectoralis minor depresses
- Stabilize the shoulder for lifting and reaching
- Serratus anterior weakness → "winged scapula"

Muscles That Move the Humerus



Muscles that move the humerus, including the rotator cuff

- Deltoid abducts; pectoralis major adducts/flexes
- Latissimus dorsi extends and medially rotates
- Rotator cuff (SITS) stabilizes the joint
- Most mobile — and least stable — joint in the body

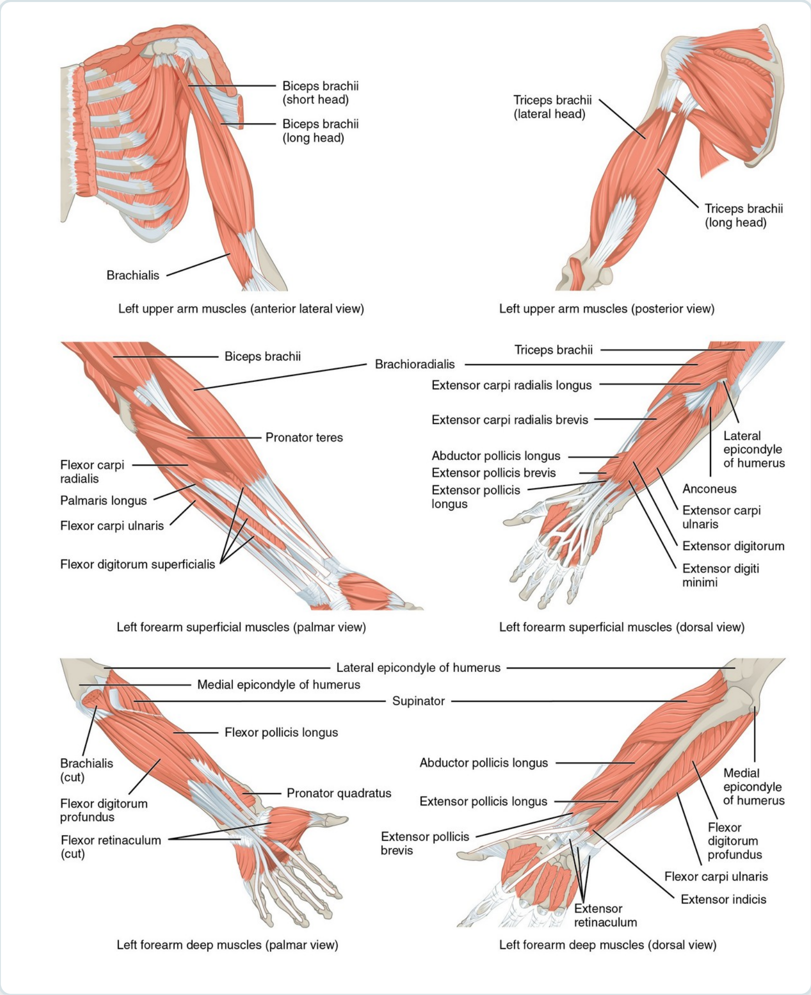
Humerus Muscles — Reference Table

Movement	Target	Target motion direction	Prime mover	Origin	Insertion
Axial muscles					
Brings elbows together; moves elbow up (as during an uppercut punch)	Humerus	Flexion; adduction; medial rotation	Pectoralis major	Clavicle; sternum; cartilage of certain ribs (1–6 or 1–7); aponeurosis of external oblique muscle	Greater tubercle of humerus
Moves elbow back (as in elbowing someone standing behind you); spreads elbows apart	Humerus; scapula	Humerus: extension, adduction, and medial rotation; scapula: depression	Latissimus dorsi	Thoracic vertebrae (T7–T12); lumbar vertebrae; lower ribs (9–12); iliac crest	Intertubercular sulcus of humerus
Scapular muscles					
Lifts arms at shoulder	Humerus	Abduction; flexion; extension; medial and lateral rotation	Deltoid	Trapezius; clavicle; acromion; spine of scapula	Deltoid tuberosity of humerus
Assists pectoralis major in bringing elbows together and stabilizes shoulder joint during movement of the pectoral girdle	Humerus	Medial rotation	Subscapularis	Subscapular fossa of scapula	Lesser tubercle of humerus
Rotates elbow outwards, as during a tennis swing	Humerus	Abduction	Supraspinatus	Supraspinous fossa of scapula	Greater tubercle of humerus
Rotates elbow outwards, as during a tennis swing	Humerus	Extension; adduction	Infraspinatus	Infraspinous fossa of scapula	Greater tubercle of humerus
Assists with medial rotation at the shoulder	Humerus	Extension; adduction	Teres major	Posterior surface of scapula	Intertubercular sulcus of humerus
Assists infraspinatus in rotating elbow outwards	Humerus	Extension; adduction	Teres minor	Lateral border of dorsal scapular surface	Greater tubercle of humerus
Moves elbow up and across body, as when putting hand on chest	Humerus	Flexion; adduction	Coracobra chialis	Coracoid process of scapula	Medial surface of humerus shaft

Muscles that move the humerus: summary table

- Sort movers by action: flex, extend, abduct
- Separate rotator cuff from the prime movers
- Link origin/insertion to the motion produced
- Use for skills lab and exam review

Muscles That Move the Forearm



Muscles of the arm that move the forearm

- Biceps brachii and brachialis flex the elbow
- Triceps brachii extends the elbow
- Supinator and pronators rotate the forearm
- Anterior vs. posterior compartments group them

Forearm — Muscles of the Wrist (Table)

Movement	Target	Target motion direction	Prime mover	Origin	Insertion
Anterior muscles (flexion)					
Performs a bicep curl; also allows palm of hand to point toward body while flexing	Forearm	Flexion; supination	Biceps brachii	Coracoid process; tubercle above glenoid cavity	Radial tuberosity
	Forearm	Flexion	Brachialis	Front of distal humerus	Coronoid process of ulna
Assists and stabilizes elbow during bicep-curl motion	Forearm	Flexion	Brachioradialis	Lateral supracondylar ridge at distal end of humerus	Base of styloid process of radius
Posterior muscles (extension)					
Extends forearm, as during a punch	Forearm	Extension	Triceps brachii	Infraglenoid tubercle of scapula; posterior shaft of humerus; posterior humeral shaft distal to radial groove	Olecranon process of ulna
Assists in extending forearm; also allows forearm to extend away from body	Forearm	Extension; abduction	Anconeus	Lateral epicondyle of humerus	Lateral aspect of olecranon process of ulna
Anterior muscles (pronation)					
Turns hand palm-down	Forearm	Pronation	Pronator teres	Medial epicondyle of humerus; coronoid process of ulna	Lateral radius
Assists in turning hand palm-down	Forearm	Pronation	Pronator quadratus	Distal portion of anterior ulnar shaft	Distal surface of anterior radius
Posterior muscles (supination)					
Turns hand palm-up	Forearm	Supination	Supinator	Lateral epicondyle of humerus; proximal ulna	Proximal end of radius

Muscles that move the forearm: reference table

- Anterior compartment flexes wrist and fingers
- Posterior compartment extends wrist and fingers
- Long tendons cross from forearm into the hand
- Median, ulnar, and radial nerve supply

Wrist & Hand — Reference Table

Movement	Target	Target motion direction	Prime mover	Origin	Insertion
Superficial anterior compartment of forearm					
Bends wrist toward body; tilts hand to side away from body	Wrist; hand	Flexion; abduction	Flexor carpi radialis	Medial epicondyle of humerus	Base of second and third metacarpals
Assists in bending hand up toward shoulder	Wrist	Flexion	Palmaris longus	Medial epicondyle of humerus	Palmar aponeurosis; skin and fascia of palm
Assists in bending hand up toward shoulder; tilts hand to side away from body; stabilizes wrist	Wrist; hand	Flexion, adduction	Flexor carpi ulnaris	Medial epicondyle of humerus; olecranon process; posterior surface of ulna	Pisiform, hamate bones, and base of fifth metacarpal
Bends fingers to make a fist	Wrist; fingers 2–5	Flexion	Flexor digitorum superficialis	Medial epicondyle of humerus; coronoid process of ulna; shaft of radius	Middle phalanges of fingers 2–5
Deep anterior compartment of forearm					
Bends tip of thumb	Thumb	Flexion	Flexor pollicis longus	Anterior surface of radius; interosseous membrane	Distal phalanx of thumb
Bends fingers to make a fist; also bends wrist toward body	Wrist; fingers	Flexion	Flexor digitorum profundus	Coronoid process; anteromedial surface of ulna; interosseous membrane	Distal phalanges of fingers 2–5
Superficial posterior compartment of forearm					
Straightens wrist away from body; tilts hand to side away from body	Wrist	Extension; abduction	Extensor radialis longus	Lateral supracondylar ridge of humerus	Base of second metacarpal
Assists extensor radialis longus in extending and abducting wrist; also stabilizes hand during finger flexion.	Wrist	Extension, abduction	Extensor carpi radialis brevis	Lateral epicondyle of humerus	Base of third metacarpal
Opens fingers and moves them sideways away from the body	Wrist; fingers	Extension; abduction	Extensor digitorum	Lateral epicondyle of humerus	Extensor expansions; distal phalanges of fingers
Extends little finger	Little finger	Extension	Extensor digiti minimi	Lateral epicondyle of humerus	Extensor expansion; distal phalanx of finger 5
Straightens wrist away from body; tilts hand to side toward body	Wrist	Extension; adduction	Extensor carpi ulnaris	Lateral epicondyle of humerus; posterior border of ulna	Base of fifth metacarpal
Deep posterior compartment of forearm					
Moves thumb sideways toward body; extends thumb; moves hand sideways toward body	Wrist; thumb	Thumb: abduction, extension; wrist: abduction	Abductor pollicis longus	Posterior surface of radius and ulna; interosseous membrane	Base of first metacarpal; trapezium
Extends thumb	Thumb	Extension	Extensor pollicis brevis	Dorsal shaft of radius and ulna; interosseous membrane	Base of proximal phalanx of thumb
Extends thumb	Thumb	Extension	Extensor pollicis longus	Dorsal shaft of radius and ulna; interosseous membrane	Base of distal phalanx of thumb
Extends index finger; straightens wrist away from body	Wrist; index finger	Extension	Extensor indicis	Posterior surface of distal ulna; interosseous membrane	Tendon of extensor digitorum of index finger

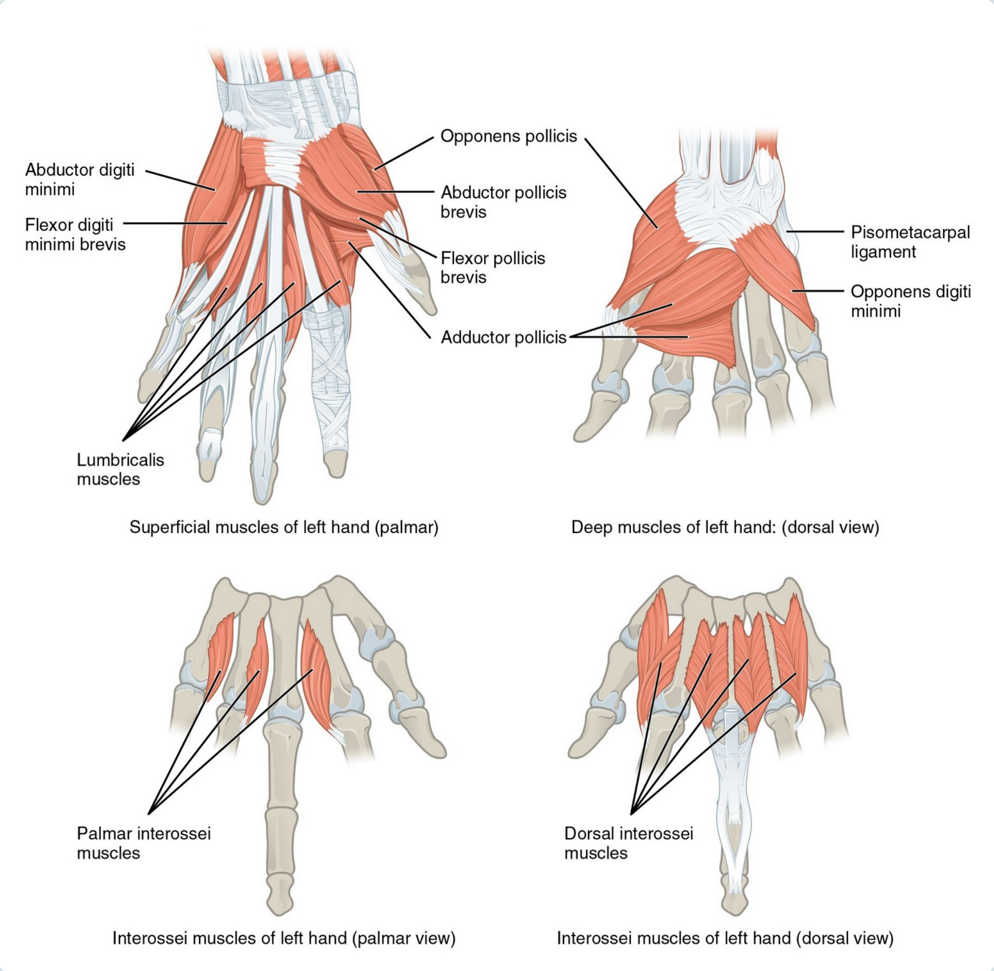
Muscles of the wrist and hand: reference table

- Extrinsic muscles power gross grip
- Retinacula stabilize the long tendons
- Carpal tunnel transmits the median nerve + tendons
- Clinical: carpal tunnel syndrome

SECTION 11.5

Intrinsic Muscles of the Hand

- Thenar, hypothenar, and midpalmar groups
- Provide fine, precise finger movements
- Thumb opposition enables a functional grasp
- Nerve injury impairs dexterity and grip



Intrinsic muscles of the hand

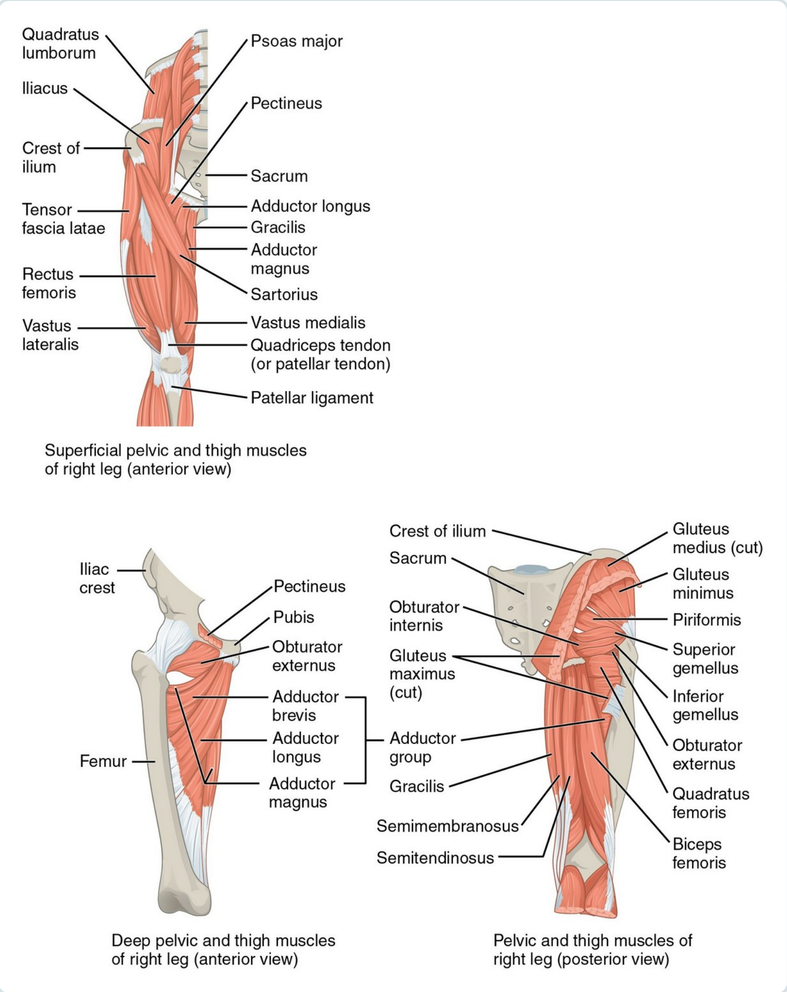
11.6

SECTION

Appendicular Muscles of the Pelvic Girdle & Lower Limbs

The hip, thigh, leg, and foot muscles that power standing, walking, and balance.

Muscles of the Gluteal Region & Thigh



Muscles of the gluteal region acting on the femur

- Gluteus maximus extends the hip
- Gluteus medius/minimus abduct and stabilize
- Iliopsoas is the main hip flexor
- Gait and balance depend on these muscles

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Gluteal Region — Reference Table

Movement	Target	Target motion direction	Prime mover	Origin	Insertion
Iliopsoas group					
Raises knee at hip, as if performing a knee attack; assists lateral rotators in twisting thigh (and lower leg) outward; assists with bending over, maintaining posture	Femur	Thigh: flexion and lateral rotation; torso: flexion	Psoas major	Lumbar vertebrae (L1–L5); thoracic vertebra (T12)	Lesser trochanter of femur
Raises knee at hip, as if performing a knee attack; assists lateral rotators in twisting thigh (and lower leg) outward; assists with bending over, maintaining posture	Femur	Thigh: flexion and lateral rotation; torso: flexion	Iliacus	Iliac fossa; iliac crest; lateral sacrum	Lesser trochanter of femur
Gluteal group					
Lowers knee and moves thigh back, as when getting ready to kick a ball	Femur	Extension	Gluteus maximus	Dorsal ilium; sacrum; coccyx	Gluteal tuberosity of femur; iliotibial tract
Opens thighs, as when doing a split	Femur	Abduction	Gluteus medius	Lateral surface of ilium	Greater trochanter of femur
Brings the thighs back together	Femur	Abduction	Gluteus minimus	External surface of ilium	Greater trochanter of femur
Assists with raising knee at hip and opening thighs; maintains posture by stabilizing the iliotibial track, which connects to the knee	Femur	Flexion; abduction	Tensor fascia lata	Anterior aspect of iliac crest; anterior superior iliac spine	Iliotibial tract
Lateral rotators					
Twists thigh (and lower leg) outward; maintains posture by stabilizing hip joint	Femur	Lateral rotation	Piriformis	Anterolateral surface of sacrum	Greater trochanter of femur
Twists thigh (and lower leg) outward; maintains posture by stabilizing hip joint	Femur	Lateral rotation	Obutrator internus	Inner surface of obturator membrane; greater sciatic notch; margins of obturator foramen	Greater trochanter in front of piriformis
Twists thigh (and lower leg) outward; maintains posture by stabilizing hip joint	Femur	Lateral rotation	Obturator externus	Outer surfaces of obturator membrane, pubic, and ischium; margins of obturator foramen	Trochanteric fossa of posterior femur
Twists thigh (and lower leg) outward; maintains posture by stabilizing hip joint	Femur	Lateral rotation	Superior gemellus	Ischial spine	Greater trochanter of femur
Twists thigh (and lower leg) outward; maintains posture by stabilizing hip joint	Femur	Lateral rotation	Inferior gemellus	Ischial tuberosity	Greater trochanter of femur
Twists thigh (and lower leg) outward; maintains posture by stabilizing hip joint	Femur	Lateral rotation	Quadratus femoris	Ischial tuberosity	Trochanteric crest of femur
Adductors					
Brings the thighs back together; assists with raising the knee	Femur	Adduction; flexion	Adductor longus	Pubis near pubic symphysis	Linea aspera
Brings the thighs back together; assists with raising the knee	Femur	Adduction; flexion	Adductor brevis	Body of pubis; inferior ramus of pubis	Linea aspera above adductor longus
Brings the thighs back together; assists with raising the knee and moving the thigh back	Femur	Adduction; flexion; extension	Adductor magnus	Ischial rami; pubic rami; ischial tuberosity	Linea aspera; adductor tubercle of femur
Opens thighs; assists with raising the knee and turning the thigh (and lower leg) inward	Femur	Adduction; flexion; medial rotation	Pectineus	Pectineal line of pubis	Lesser trochanter to linea aspera of posterior aspect of femur

Muscles of the gluteal region: summary table

- Separate flexors, extensors, abductors, rotators
- Note the pelvic stabilizers used in gait
- Connect to fall risk and mobility
- Reference for clinical skills

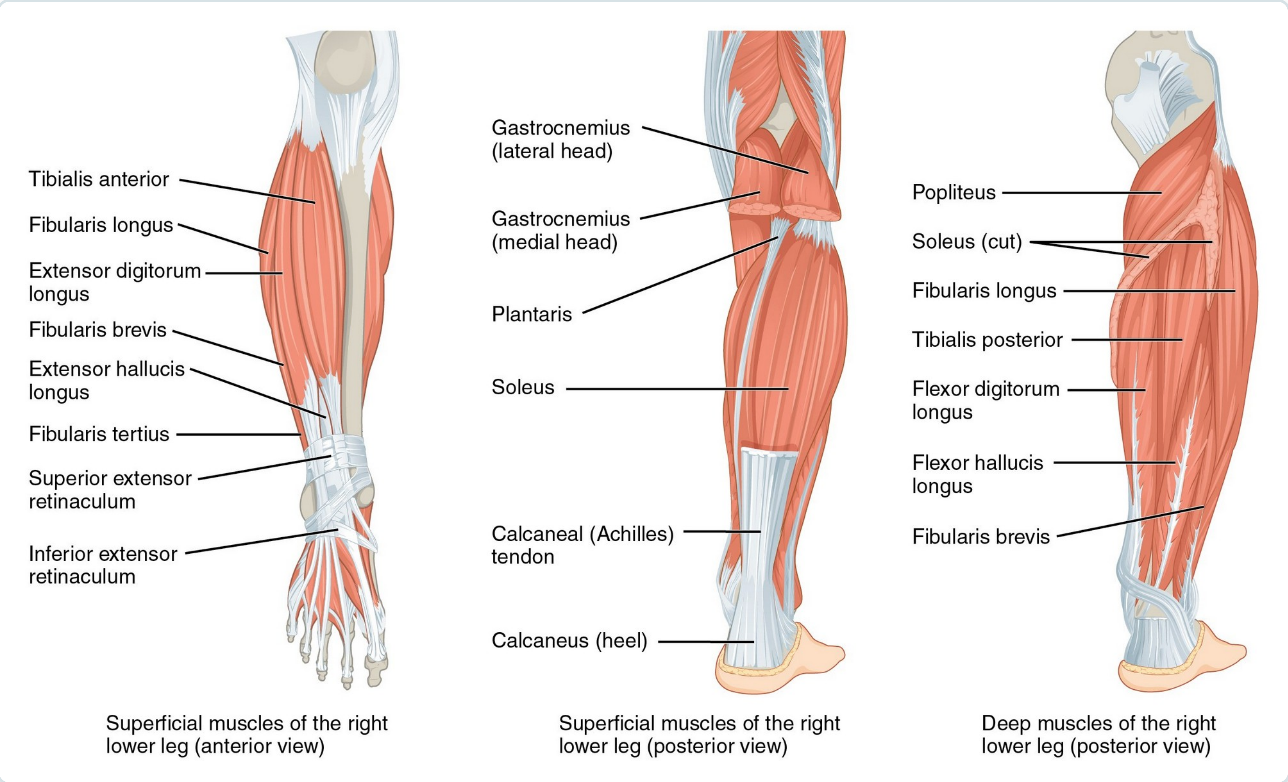
Muscles of the Thigh — Reference Table

Movement	Target	Target motion direction	Prime mover	Origin	Insertion
Medial compartment of thigh					
Moves back of lower legs up toward buttocks, as when kneeling; assists in opening thighs	Femur; tibia/fibula	Tibia/fibula: flexion; thigh: adduction	Gracilis	Inferior ramus; body of pubis; ischial ramus	Medial surface of tibia
Anterior compartment of thigh: Quadriceps femoris group					
Moves lower leg out in front of body, as when kicking; assists in raising the knee	Femur; tibia/fibula	Tibia/fibula: extension; thigh: flexion	Rectus femoris	Anterior inferior iliac spine; superior margin of acetabulum	Patella; tibial tuberosity
Moves lower leg out in front of body, as when kicking	Tibia/fibula	Extension	Vastus lateralis	Greater trochanter; intertrochanteric line; linea aspera	Patella; tibial tuberosity
Moves lower leg out in front of body, as when kicking	Tibia/fibula	Extension	Vastus medialis	Linea aspera; intertrochanteric line	Patella; tibial tuberosity
Moves lower leg out in front of body, as when kicking	Tibia/fibula	Extension	Vastus intermedius	Proximal femur shaft	Patella; tibial tuberosity
Moves back of lower legs up and back toward the buttocks, as when kneeling; assists in moving thigh diagonally upward and outward as when mounting a bike	Femur; tibia/fibula	Tibia: flexion; thigh: flexion, abduction, lateral rotation	Sartorius	Anterior superior iliac spine	Medial aspect of proximal tibia
Posterior compartment of thigh: Hamstring group					
Moves back of lower legs up and back toward the buttocks, as when kneeling; moves thigh down and back; twists the thigh (and lower leg) outward	Femur; tibia/fibula	Tibia/fibula: flexion; thigh: extension, lateral rotation	Biceps femoris	Ischial tuberosity; linea aspera; distal femur	Head of fibula; lateral condyle of tibia
Moves back of lower legs up toward buttocks, as when kneeling; moves thigh down and back; twists the thigh (and lower leg) inward	Femur; tibia/fibula	Tibia/fibula: flexion; thigh: extension, medial rotation	Semitendinosus	Ischial tuberosity	Upper tibial shaft
Moves back of lower legs up and back toward the buttocks as when kneeling; moves thigh down and back; twists the thigh (and lower leg) inward	Femur; tibia/fibula	Tibia/fibula: flexion; thigh: extension, medial rotation	Semi-membranosus	Ischial tuberosity	Medial condyle of tibia; lateral condyle of femur

Muscles of the thigh: reference table

- Quadriceps extend the knee
- Hamstrings flex the knee and extend the hip
- Adductors pull the thigh medially
- Powerful muscles for transfers and walking

Muscles That Move the Foot



Muscles of the leg that move the foot

- Anterior leg dorsiflexes (tibialis anterior)
- Posterior leg plantarflexes (gastrocnemius, soleus)
- Calf muscles form the Achilles tendon
- Deep fibular nerve injury → "foot drop"

SECTION 11.6

Leg Muscles — Reference Table

Movement	Target	Target motion direction	Prime mover	Origin	Insertion
Anterior compartment of leg					
Raises the sole of the foot off the ground, as when preparing to foot-tap; bends the inside of the foot upwards, as when catching your balance while falling laterally toward the opposite side as the balancing foot	Foot	Dorsiflexion; inversion	Tibialis anterior	Lateral condyle and upper tibial shaft; interosseous membrane	Interior surface of medial cuneiform; First metatarsal bone
Raises the sole of the foot off the ground, as when preparing to foot-tap; extends the big toe	Foot; big toe	Foot: dorsiflexion; big toe: extension	Extensor hallucis longus	Anteromedial fibula shaft; interosseous membrane	Distal phalanx of big toe
Raises the sole of the foot off the ground, as when preparing to foot-tap; extends toes	Foot; toes 2–5	Foot: dorsiflexion; toes: extension	Extensor digitorum longus	Lateral condyle of tibia; proximal portion of fibula; interosseous membrane	Middle and distal phalanges of toes 2–5
Lateral compartment of leg					
Lowers the sole of the foot to the ground, as when foot-tapping or jumping; bends the inside of the foot downwards, as when catching your balance while falling laterally toward the same side as the balancing foot	Foot	Plantar flexion and eversion	Fibularis longus	Upper portion of lateral fibula	First metatarsal; medial cuneiform
Lowers the sole of the foot to the ground, as when foot-tapping or jumping; bends the inside of the foot downward, as when catching your balance while falling laterally toward the same side as the balancing foot	Foot	Plantar flexion and eversion	Fibularis (peroneus) brevis	Distal fibula shaft	Proximal end of fifth metatarsal
Posterior compartment of leg: Superficial muscles					
Lowers the sole of the foot to the ground, as when foot-tapping or jumping; assists in moving the back of the lower legs up and back toward the buttocks	Foot; tibia/ fibula	Foot: plantar flexion; tibia/fibula: flexion	Gastrocnemius	Medial and lateral condyles of femur	Posterior calcaneus
Lowers the sole of the foot to the ground, as when foot-tapping or jumping; maintains posture while walking	Foot	Plantar flexion	Soleus	Superior tibia; fibula; interosseous membrane	Posterior calcaneus
Lowers the sole of the foot to the ground, as when foot-tapping or jumping; assists in moving the back of the lower legs up and back toward the buttocks	Foot; tibia/ fibula	Foot: plantar flexion; tibia/fibula: flexion	Plantaris	Posterior femur above lateral condyle	Calcaneus or calcaneus tendon
Lowers the sole of the foot to the ground, as when foot-tapping or jumping	Foot	Plantar flexion	Tibialis posterior	Superior tibia and fibula; interosseous membrane	Several tarsals and metatarsals 2–4
Posterior compartment of leg: Deep muscles					
Moves the back of the lower legs up and back toward the buttocks; assists in rotation of the leg at the knee and thigh	Tibia/ fibula	Tibia/fibula: flexion thigh and lower leg: medial and lateral rotation	Popliteus	Lateral condyle of femur; lateral meniscus	Proximal tibia
Lowers the sole of the foot to the ground, as when foot-tapping or jumping; bends the inside of the foot upward and flexes toes	Foot; toes 2–5	Foot: plantar flexion and inversion toes: flexion	Flexor digitorum longus	Posterior tibia	Distal phalanges of toes 2–5
Flexes the big toe	Big toe; foot	Big toe: flexion foot: plantar flexion	Flexor hallucis longus	Midshaft of fibula; interosseous membrane	Distal phalanx of big toe

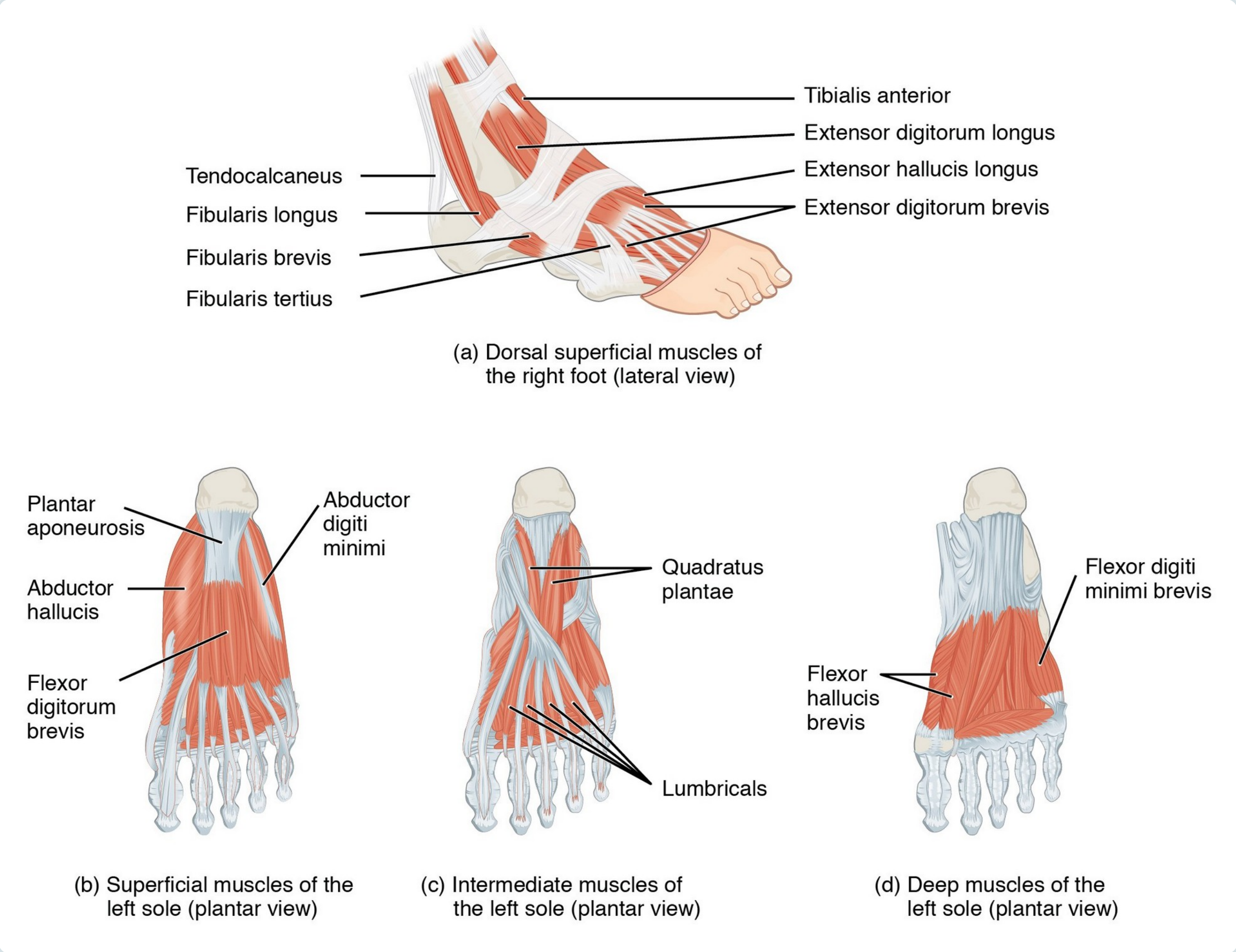
Muscles of the leg: reference table

- Group by compartment: anterior, lateral, posterior
- Match each muscle to its foot movement
- Note tendons crossing the ankle
- Compartment syndrome awareness

SECTION 11.6

Intrinsic Muscles of the Foot

- Support the arches of the foot
- Provide balance and fine foot control
- Layered plantar muscles cushion body weight
- Arch support matters in diabetic foot care



Intrinsic muscles of the foot

SECTION 11.6

Foot Muscles — Reference Table

Movement	Target	Target motion direction	Prime mover	Origin	Insertion
Dorsal group					
Extends toes 2–5	Toes 2–5	Extension	Extensor digitorum brevis	Calcaneus; extensor retinaculum	Base of proximal phalanx of big toe; extensor expansions on toes 2–5
Plantar group (layer 1)					
Abducts and flexes big toe	Big toe	Adduction; flexion	Abductor hallucis	Calcaneal tuberosity; flexor retinaculum	Proximal phalanx of big toe
Flexes toes 2–4	Middle toes	Flexion	Flexor digitorum brevis	Calcaneal tuberosity	Middle phalanx of toes 2–4
Abducts and flexes small toe	Toe 5	Abduction; flexion	Abductor digiti minimi	Calcaneal tuberosity	Proximal phalanx of little toe
Plantar group (layer 2)					
Assists in flexing toes 2–5	Toes 2–5	Flexion	Quadratus plantae	Medial and lateral sides of calcaneus	Tendon of flexor digitorum longus
Extends toes 2–5 at the interphalangeal joints; flexes the small toes at the metatarsophalangeal joints	Toes 2–5	Extension; flexion	Lumbricals	Tendons of flexor digitorum longus	Medial side of proximal phalanx of toes 2–5
Plantar group (layer 3)					
Flexes big toe	Big toe	Flexion	Flexor hallucis brevis	Lateral cuneiform; cuboid bones	Base of proximal phalanx of big toe
Adducts and flexes big toe	Big toe	Adduction; flexion	Adductor hallucis	Bases of metatarsals 2–4; fibularis longus tendon sheath; ligament across metatarsophalangeal joints	Base of proximal phalanx of big toe
Flexes small toe	Little toe	Flexion	Flexor digiti minimi brevis	Base of metatarsal 5; tendon sheath of fibularis longus	Base of proximal phalanx of little toe
Plantar group (layer 4)					
Abducts and flexes middle toes at metatarsophalangeal joints; extends middle toes at interphalangeal joints	Middle toes	Abduction; flexion; extension	Dorsal interossei	Sides of metatarsals	Both sides of toe 2; for each other toe, extensor expansion over first phalanx on side opposite toe 2
Abducts toes 3–5; flexes proximal phalanges and extends distal phalanges	Small toes	Abduction; flexion; extension	Plantar interossei	Side of each metatarsal that faces metatarsal 2 (absent from metatarsal 2)	Extensor expansion on first phalanx of each toe (except to 2) on side facing toe 2

Muscles of the foot: reference table

- Plantar muscles arranged in four layers
- Work with intrinsic and extrinsic tendons
- Maintain the arches and move the toes
- Reference for foot assessment

CHAPTER 11

Key Takeaways

- Muscles act in coordinated groups — agonist, antagonist, synergist, and fixator.
- Fascicle arrangement sets the trade-off between force and range of motion.
- Muscle names encode location, shape, size, direction, action, and attachments.
- Axial muscles support the head, neck, trunk, breathing, and continence.
- Appendicular muscles move the limbs and underlie gait, grasp, and balance.

CLINICAL CONNECTIONS

Why the Muscular System Matters in Practical Nursing

Breathing

Diaphragm (C3–C5), intercostals, and accessory muscles drive ventilation — watch for retractions.

Mobility & Transfers

Gluteals, quadriceps, and hamstrings power standing and walking — central to fall risk and safe lifting.

Swallowing

Tongue, suprahyoid, and pharyngeal muscles protect the airway — basis for aspiration precautions.

Continence

The pelvic floor (levator ani) supports organs and controls elimination — weakness causes incontinence.

Cranial Nerves

Link muscle groups to CN V (chewing), VII (face), XII (tongue), and XI (SCM/trapezius).

Body Mechanics

Erector spinae and core muscles protect the back — reinforce lifting technique with every patient.